Predictive Maintenance of Physical Assets:

Application to Wind Turbines

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Predictive Maintenance: Importance & Goals

The **objective** of this project:

- Implement emerging AI technology into existing systems for predictive maintenance of renewable energy sources.
- The main focus of this project is on wind turbines.

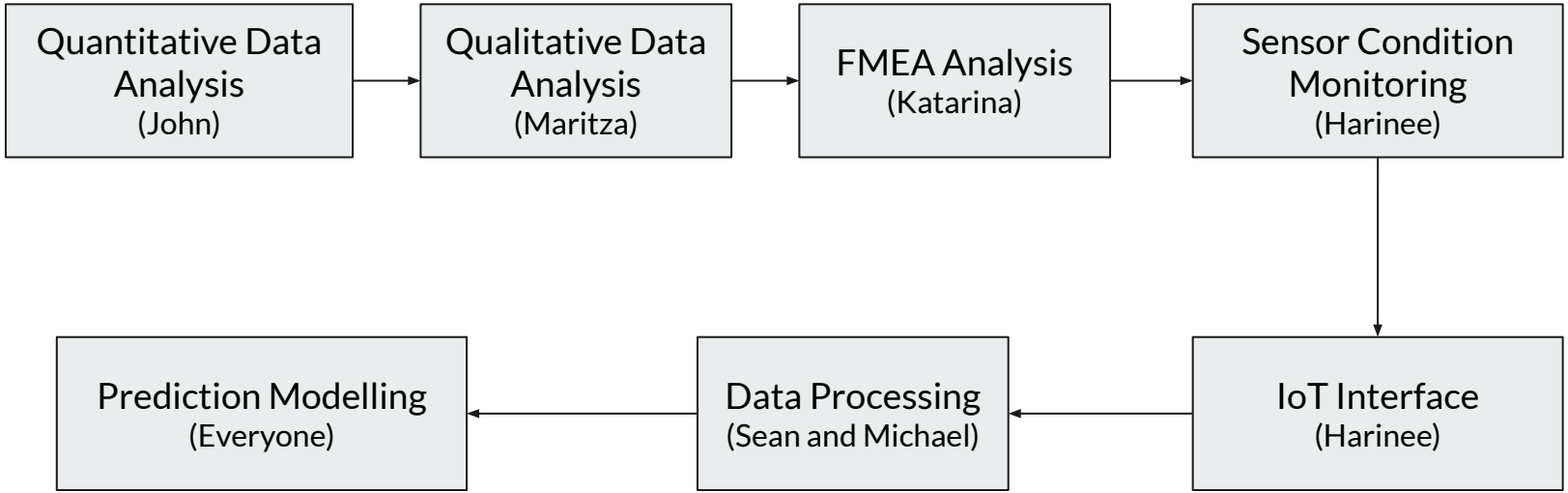
Functions & Benefits:

- Performs condition-based monitoring through continuous health assessments.
- Enhances preventative maintenance by delivering real-time insights into equipment condition.
    - Reduces operational costs by detecting early wear and damage in critical systems.
- Uses SCADA data, condition-monitoring sensors, and IoT modeling for accurate predictions.
- Extends the operational lifespan of individual wind turbines and entire wind farms.

Predictive Maintenance vs. Preventive Maintenance

Preventive Maintenance	Predictive Maintenance
Performed on a fixed schedule (time-based or usage-based).	Performed based on real-time condition and health.
Replace and service parts regardless if needed.	Service components when data indicates wear and risk for failure.
Relies on routine inspections; more unnecessary maintenance	Relies on data (SCADA, sensors, AI analytics); targets specific issues
Lower initial cost, but increased long-term operational costs	Increased initial cost, but lower long-term costs
Does not adapt to changing conditions	Able to adapt dynamically to real-time equipment data

Workflow

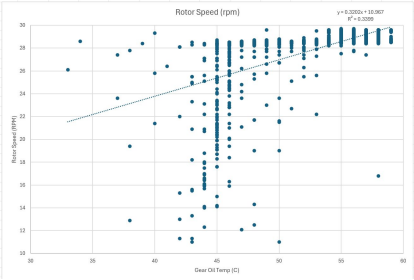

Quantitative Data Analysis
(John)
Qualitative Data Analysis
(Maritza)
FMEA Analysis
(Katarina)
Sensor Condition Monitoring
(Harinee)
IoT Interface
(Harinee)
Data Processing
(Sean and Michael)
Prediction Modelling
(Everyone)

Quantitative Analysis

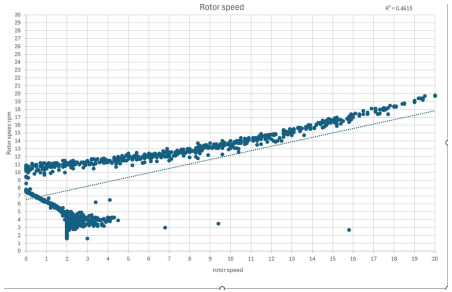
- Quantitative Data Analysis (Sikorski, John)
  - Reliability Based Analysis
  - Deriving key relationships and equations
  - Identifying qualitative factors that contribute to failure trends
  - Data cleaning
  - Scata data failure indicators ie. limit thresholds

GS Phase 3 (V)	Total Active Energy (W)	Total Reactive Energy (W)	Hydraulic Temperature (C)	Temperature Outside Nacelle (C)	Temperature Inside Nacelle	Gearbox Oil Temperature (C)	Generator Stator Temperature (C)	Slip Ring Temperature (C)	Controller Temperature (C)	Active Power Generation (W)	Active Power Generation (W)	Rotor Speed (rpm)	Rotor Speed (RS)	Generator Speed (rpm)	Generator Speed (rpm).1	Generator Speed (rpm).2	Pitch Angle for Blade 1 (deg)	Pitch Angle for Blade 1 (deg)	Wind Speed (m/s)	Wind Speed Outside Nacelle (m/s)	Wind Speed Outside Nacelle (m/s)	delta_s
407.4	49182	-2189	24	2	16	58	61		25	294400	398900	28.7	28.8	1515	1522	1508	0	0.9	7.8	9.5	7.8	600
407.4	64491	-6445	24	2	16	55	59		25	387800	517100	28.8	28.9	1520	1528	1514	-0.7	0.3	8.6	10.3	8.6	600
408.5	52259	-3042	24	2	15	56	50		25	314400	473800	28.7	29	1519	1529	1514	-0.1	0.6	7.9	9.7	7.9	600
411.8	109972	-23896	36	3	23	57	102		31	659800	663000	29.5	30.5	1561	1615	1531	8.1	11.8	12.7	15	12.7	600
412.5	109982	-23848	37	3	23	55	104		31	659900	662800	29.6	30.6	1563	1619	1522	9.5	14.9	13.4	18.3	13.4	600
408.2	109973	-24135	37	2	22	57	105		31	659800	672000	29.4	30.2	1556	1599	1518	9.5	13.3	13.4	16.9	13.4	600
406.6	109975	-24174	37	2	22	57	106		31	659900	663300	29.4	30.7	1557	1621	1512	13.8	17.4	15.8	19.8	15.8	600
406.7	109978	-24266	37	3	22	55	107		31	659900	675600	29.5	31	1562	1643	1515	19.5	23.2	19.7	23.5	19.7	600
406.5	109984	-24331	37	2	23	58	108		31	659900	669100	29.5	30.8	1562	1633	1513	14.6	20.9	15.8	22.7	15.8	600
407.2	109975	-24463	37	1	23	58	109		30	659800	686100	29.5	30.9	1562	1636	1511	21.7	25.1	20.5	25.1	20.5	600
405.5	109987	-24542	37	1	21	57	109		29	659900	674900	29.5	30.8	1559	1648	1509	21.6	26.1	20.4	25.1	20.4	600
404.5	109980	-24351	37	1	20	57	110		29	659900	690800	29.5	30.9	1561	1637	1512	21.8	26.8	20.5	26.2	20.5	600
407	0	0	17	-1	4	31	42		16	-500	-300	0.5	2	68	152	11	83	86.7	25.6	35.3	25.6	16200
407.6	0	0	9	1	5	16	19		16	-600	-300	0.2	1.6	60	109	13	83	83	19.9	26	19.9	600
406	0	0	9	1	6	14	18		16	-300	-300	0	1.6	51	97	11	83	83	16	20.9	16	600
405.3	0	0	9	2	6	13	16		17	-300	-300	0	0	41	71	13	83	83	11.6	15.7	11.6	600
405.9	108100	-23438	11	3	12	40	52		18	648700	680700	29.3	30.2	1551	1596	1518	2.1	8.1	11	13.3	11	600
403.6	106564	-23070	21	4	19	58	87		23	639400	680800	29.2	30.3	1545	1599	1517	1.2	9	10.7	12.8	10.7	600
403.4	101351	-20669	28	5	22	56	98		28	608500	680600	29.1	30.1	1540	1593	1515	0.3	7.4	10.6	13.1	10.6	600
404.3	105913	-22733	31	5	24	55	100		31	635500	681400	29.3	30.2	1550	1598	1522	1.4	7.1	11	13.3	11	600
403.6	109974	-24602	35	6	25	54	107		33	659800	668000	29.5	30.4	1562	1606	1529	6.3	12.7	12.3	16.3	12.3	600
404.5	109984	-24569	39	5	26	55	111		35	659900	663900	29.5	30.6	1560	1628	1528	8.3	13.4	12.5	15.3	12.5	600
409.3	109980	-24246	41	5	26	55	111		34	659900	663700	29.5	30.4	1562	1606	1533	8.6	13.5	13.2	16.6	13.2	600
409.8	109977	-24268	42	4	24	56	112		34	659900	663800	29.5	30.5	1562	1610	1520	11.8	16.4	14.7	18.4	14.7	600
408.6	109971	-24113	42	4	23	56	111		33	659800	663500	29.5	30.5	1561	1615	1522	11.7	16	14.6	18.2	14.6	600
408.1	109976	-24317	42	4	24	55	111		33	659800	663500	29.5	30.6	1560	1622	1524	13.1	17.8	15.5	19.3	15.5	600
408.7	109976	-24029	42	4	21	56	111		33	659800	663200	29.5	30.3	1561	1603	1522	15.1	18.9	16.9	20.2	16.9	600
408.6	109978	-24202	42	4	23	56	112		33	659900	698300	29.6	30.9	1563	1644	1512	14.2	18.2	16	19.8	16	600
407.8	109976	-24004	43	4	23	57	111		33	659900	681000	29.6	31.2	1564	1649	1518	10.7	17.3	13.6	18	13.6	600
408.1	98019	-19170	31	4	21	56	81	205	29	587500	679900	29.1	30.3	1538	1595	1516	0	8.1	10.2	12.9	10.2	600
407.8	109634	-24274	33	5	22	57	89	205	31	656200	679800	29.5	30.4	1558	1609	1521	3.9	10.2	11.6	14.3	11.6	600
408.2	106296	-23848	36	6	24	54	98	205	33	636200	681700	29.4	30.5	1556	1618	1516	3.3	11.3	11	13.8	11	1200

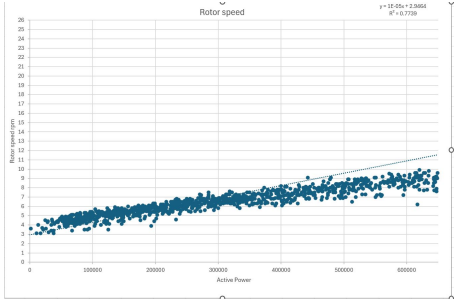
Graphs and Equations



Rotor Speed vs. Generator Temp



Rotor Speed vs. Blade Pitch



Rotor Speed vs. Active Power

Next steps: Relationship Categorization Matlab Interpretation and Solve

In order to better understand where qualitative factors are contributing factors:

- Process of Matlab to match qualitative instances with failure indications

- Derived systems for variables solved and output a qualitative failure indicator

Qualitative Data Analysis Approach

How the predictive maintenance insights were developed

Data sources (Qualitative)

- Technical Failure Descriptions for major wind turbine components
- Maintenance logs, root cause summaries, and probability of failure data
- Preventive maintenance recommendations from OEM procedures
- Field behavior patterns (vibration alarms, corrosion indicators, hydraulic leakage trends)

Coding & Thematic Grouping

I grouped all qualitative findings into recurring themes

- *Wear- and fatigue-related failures* (cracking, pitting, delamination)
- *Contamination-related failures* (oil contamination, moisture ingress, dust)
- *Control-system degradation* (sensor drift, encoder failure, actuator malfunction)
- *Environmental stress* (lightning strikes, corrosion in humid/coastal sites)

Pattern Identification

Across components, patterns showed that failures consistently cluster around:

- Loss of lubrication or oil quality
- Improper torque/preload
- Environmental exposure (humidity, salt, lightning)
- Cyclic mechanical stress
- Aging of seals, hoses, and sensors

Identified failures

Rotor System	Fatigue cracks, erosion, Bolt loosening, Lightning damage, Pitch mechanism failure, Bearing failure, Misalignment, Sensor drift
Nacelle / Drive Train	Gearbox overheating, Oil leaks, Pump failure, Motor failure, Generator bearing wear, Cooling system failure
Electrical Systems	IGBT failure, Capacitor aging, Breaker trip, Insulation degradation, Software or communication error, Module logic failure, Battery degradation
Control & Communication Systems	Feedback sensor fault, Signal loss, Communication loss, Software faults
Tower & Structural Components	Corrosion, Structural cracking, Foundation settlement

Component	Imapcated how?	Conspequnce
Blades	Erosion, cracking, lightning damage	Loss of efficiency, catastrophic detachment
Pitch System	Icing, actuator failure	Overspeed → turbine shutdown, safety risk
Gearbox	Wear, overheating, lubrication failure	Major downtime; most expensive repair
Generator	Electrical overheating, bearing failure	Power production loss, costly replacement
Power Electronics (IGBTs)	Thermal stress, capacitor aging	Total system shutdown; grid instability
Sensors & Communication	Drift, signal loss	Loss of control accuracy → turbine derates or halts
Tower / Structure	Corrison, cracks	Tower collapse in extreme cases

#1 Rotor System – Highest number of distinct failures

Why this system is most failure prone?

The rotor is constantly exposed to mechanical loads, weather, and high-speed rotation, increasing wear, fatigue, and environmental damage.

Which components are most impacted?

If These Systems Fail, What Happens?

Rotor System Failure

- Massive drop in power production
- Risk of overspeed and catastrophic blade failure
- Possible emergency shutdown

The rotor captures the wind if it fails, the turbine produces zero power.

Gearbox Failure

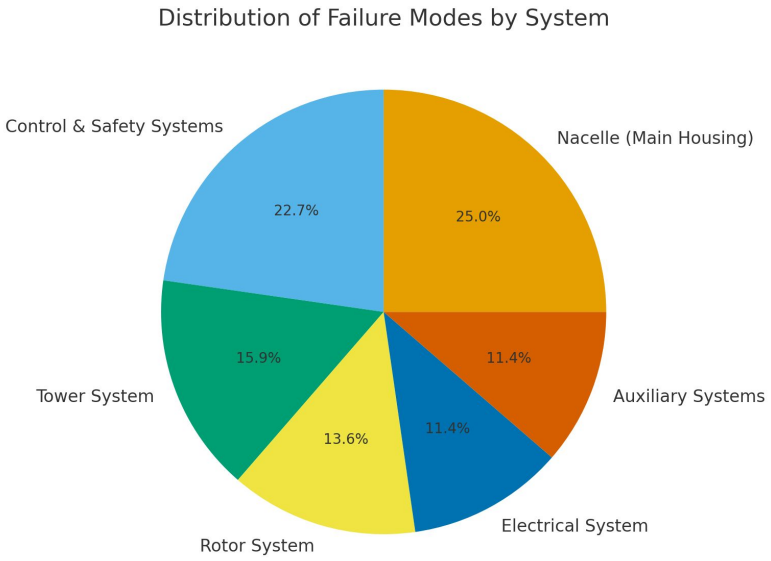
- Entire turbine stops
- Multi-day repair downtime
- High-cost component replacement

Gearboxes are known as the #1 cost driver in turbine failures.

Control System Failure

- Turbine may enter safe shutdown mode
- Wrong pitch angle → rotor overspeed
- Incorrect yaw → massive power losses

Controls ensure safe and optimal operation.



NEXT STEPS

1

Identify a baseline “Normal Behavior”

Predictive maintenance only works when the system knows what *normal* looks like

2

Define Early Warning indicators

For each failure mode, identify measurable early symptoms.

3

Data Monitoring and Trend Analysis

Detect failures weeks/months before they become severe.

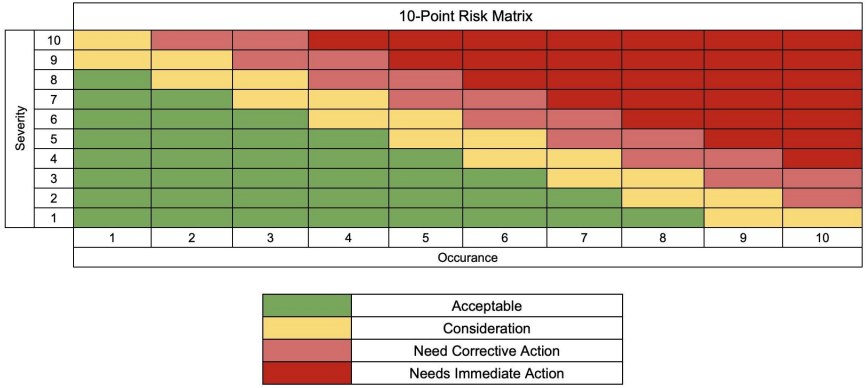
4

Connect Predictions to Maintenance Actions

Ensure predictions improve operational productivity and reduce failures.

FMEA Database and Analysis

- FMEA stands for Failure Mode and Effects Analysis and it aids in identifying and prioritizing which components require immediate attention.
- By using the root cause analysis and probability of data found in the Qualitative Data Analysis, we can rank each component, in a 10 point risk matrix based on the probability of failure and the impact of the failure. This leads to determining the effects on the overall wind turbine.



Severity	
	Minimal (1)
	Very Low (2)
	Low (3)
	Moderate (4)
	Medium (5)
	Elevated (6)
	High (7)
	Very High (8)
	Severe (9)
	Critical (10)

Occurance	
	Very Remote (1)
	Remote (2)
	Very Unlikely (3)
	Unlikely (4)
	Possible (5)
	Moderately Likely (6)
	Likely (7)
	Very Likely (8)
	Almost Certain (9)
	Inevitable (10)

Detection	
	Certain Detection (1)
	Very High (2)
	High (3)
	Moderately High (4)
	Moderate (5)
	Moderately Low (6)
	Low (7)
	Very Low (8)
	Extemely Unlikely (9)
	Impossible (10)

S*O*D = RPN (Risk Level)

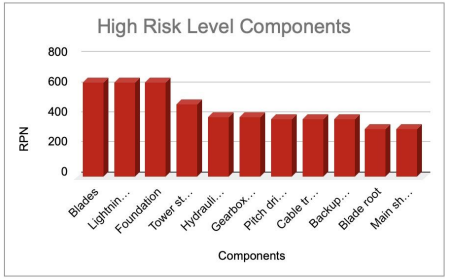
0 - 150	Low Risk
151 - 300	Medium Risk
300 +	High Risk

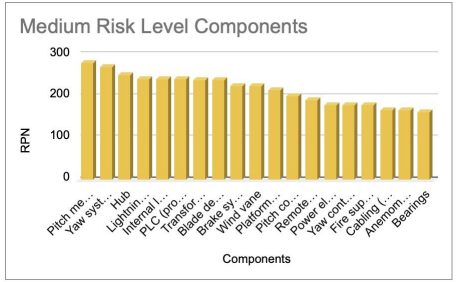
Wind Turbine Components

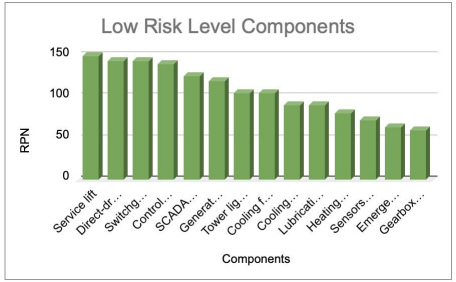
Complete FMEA Analysis of Wind Turbine Components

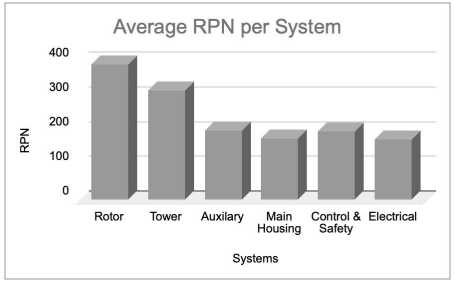
- Ranking of Severity, Occurrence, and Detection used to calculate Risk Priority Number for each component.
- Based on each component's Risk Priority Number (RPN), a risk level for each failure can be determined. From there we can rank each component from highest to lowest risk.

System	Components	S	O	D	RPN	Risk Level
Rotor	Blades	10	7	9	630	High
Rotor	Lightning protection system	10	7	9	630	High
Tower	Foundation	10	7	9	630	High
Tower	Tower structure (steel, concrete, hybrid, lattice)	10	6	8	480	High
Auxilary	Hydraulic system (for pitch, yaw, braking)	10	8	5	400	High
Main Housing	Gearbox (planetary, helical stages)	10	8	5	400	High
Rotor	Pitch drives	8	6	8	384	High
Tower	Cable trays	6	8	8	384	High
Control & Safety	Backup power systems (batteries, supercapacitors)	8	8	6	384	High
Rotor	Blade root	10	4	8	320	High
Main Housing	Main shaft (low-speed shaft)	10	4	8	320	High









FMEA Analysis using SCADA Data Relationship

Based on the relationship between the rotor system and the gearbox, found through SCADA data, we have identified maintenance procedures for each component and their cost. This shows that through the use of predictive maintenance, we have reduce overall costs of repairs.

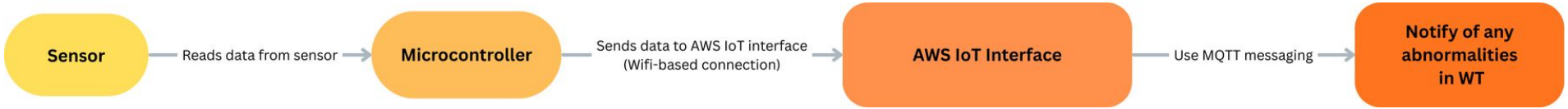
System	Components	Failure	Severity	Failure Cause	Root Cause	Occurance	Detection	RPN	Risk Level	Maintenance Procedures	Cost for Maintenance	Cost for replacement
Rotor System	Blades	Leading edge erosion	10	rain, hail, sand, and dust erode the blade's outer coating	High-speed rain/particle impact during normal operation, especially at blade tip speeds of 60–80 m/s	7	9	630	High	LEP renewal, composite repair, cleaning, erosion coating	$500–$40,000	Blade Replacement: $100,000–$250,000 per blade
	Blade root	Fatigue cracking in root bolts or laminates	10	Repeated cyclic loading from wind gusts causes micro-cracks in composite laminates or root bolts/flanges	High-cycle fatigue from turbulence, frequent start/stop cycles, and yaw misalignment	4	8	320	High	Bolt torque check, NDT, bolt replacement, laminate repair	$1,000–$40,000	Bolt Set Replacement: $8,000–$20,000
	Hub	Structural cracking of hub casting/welds	9	High cyclic bending loads from gusts or rotor imbalance produce micro-cracks in the cast or welded hub body	Casting porosity, residual welding stresses, or operating beyond design wind regime	4	7	252	Medium	NDT, weld repair, reinforcement, stress relief	$3,000–$60,000	Full Hub Replacement: $100,000–$250,000 + crane costs
	Pitch drives	Gear wear or pitting	8	Continuous oscillatory motion of drive gearbox during frequent pitch corrections	Inadequate or degraded grease/oil, misalignment of motor and ring gear, overload from abrupt braking or high turbulence	6	8	384	High	Oil change, gear inspection, gearbox replacement	$1,000–$60,000	Pitch motor replacement: $8,000–$20,000
Nacelle (Main Housing)	Gearbox (planetary, helical stages)	Gear tooth pitting / micropitting	10	High contact stress from continuous torque transmission	Inadequate lubrication film, misalignment of planetary or helical gears, contaminated oil increasing wear particles	8	5	400	High	Oil analysis, oil change, sensor installation, gear repair	$300-$40,000	Full Gearbox Replacement: $150,000–$350,000 + crane costs
	Gearbox lubrication system	Oil pump failure	4	Loss of oil flow or pressure	Pump motor burnout, bearing wear, cavitation from clogged suction strainer, low oil level, or air ingress	5	3	60	Low	Pump replacement, flush, filter change	$3,000-$8,000	Complete Replacement: $24,000

Using Sensors to Monitor Conditions

- **Condition Monitoring** can help develop preventive and predictive maintenance strategies. To achieve this, use of sensors is critical.
- A typical wind turbine has over 300 sensors, a sample of these is given in the adjoining table. These include the likes of:
  - **Vibration Sensors:** used for detecting bearing wear, imbalance, shaft misalignment
  - **Temperature Sensors:** used for detecting overheating of bearing, gearbox, main WT shaft
  - **Pressure Sensors:** used for detecting hydraulic pressure, lubrication health

Components	Sensor
Blades	Accelerometer
Blade root	Sound Sensor
Pitch mechanism (hydraulic/electric actuators, bearings)	Temperature Sensor
Lightning protection system	Voltage/Current Surge sensors
Hub	Accelerometers, Vibration Sensor, Temperature Sensor
Pitch drives	Accelerometers, Vibration Sensor, Temperature Sensor
Main shaft (low-speed shaft)	Accelerometers, Torque Sensors, Thermocouple
Bearings	Accelerometer
Gearbox (planetary, helical stages)	Accelerometer, Sound
Gearbox lubrication system	Oil-Quality sensors, temperature sensor, pressure sensor, flow meter
Direct-drive system (if applicable)	Torque sensor, temperature sensor, accelerometer/vibration sensor
Generator (SCIG, DFIG, PMSG)	Temperature sensor, vibration sensor, voltage/current sensors
Brake system (disc, hydraulic, emergency)	Pressure sensor, temperature sensor
Yaw system (motors, bearings, brakes)	Accelerometers/vibration sensors, temperature sensor, torque sensor
Cooling system (air/liquid, heat exchangers)	Temperature sensors, flow sensors, pressure sensors
Lubrication system (oil pumps, filters)	Oil-Quality sensors, temperature sensor, pressure sensor, flow meter
Sensors (vibration, temperature, strain, oil quality)	--
Tower structure (steel, concrete, hybrid, lattice)	Tilt sensors, accelerometers
Internal ladders and safety systems	Load sensors
Service lift	Emergency brake sensors
Cable trays	Temperature sensors
Platforms/rest stations	Weight/load sensors
Tower lighting (aviation compliance)	Current sensors, photocells
Foundation (gravity, pile, monopile, jacket, tripod, floating)	Accelerometers, vibration sensors
Power electronics (converters, rectifiers, inverters)	Temperature sensors, voltage/current sensors
Control boards	Temperature sensors, voltage/current sensors
Transformer (step-up, nacelle or tower base)	Temperature sensors
Switchgear (circuit breakers, disconnect switches)	Current sensors, temperature sensors
Cabling (high-voltage, control, fiber optic)	Temperature sensor
SCADA system	--
PLC (programmable logic controllers)	--
Remote monitoring & communication modules	Signal quality sensors, temperature sensors
Anemometer	Temperature sensor (ice), optical sensor
Wind vane	Potentiometer
Pitch control system	Torque/force sensors, temperature sensors
Yaw control system	Torque/force sensors, temperature sensors
Emergency stop buttons	Contact sensor + indicator that displays status
Backup power systems (batteries, supercapacitors)	Voltage sensors, current sensors, temperature sensors
Lightning protection system (receptors, down conductors, grounding)	Voltage/Current Surge sensors
Hydraulic system (for pitch, yaw, braking)	Pressure sensors, Temperature sensors, Flow sensors
Heating systems (nacelle heaters)	Temperature sensors
Blade de-icing systems (electrical, hot air)	Temperature sensors
Cooling fans and ventilation	Temperature sensors, Air-flow sensors, vibration sensors
Fire suppression system (sensors, extinguishers)	Smoke detectors, Heat detector/Temperature sensor, Pressure sensor

Sensor Condition Monitoring → IoT Interface

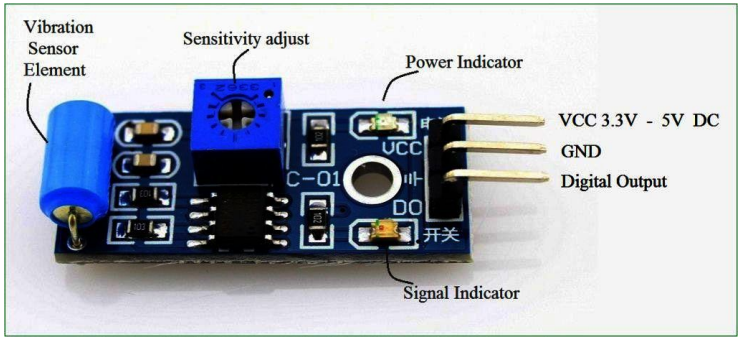


ESP32 (Microcontroller) = “Transmission Point”

- Direct interface with sensors
- Provides GPIO, ADC, and digital output
- Sends data wirelessly via Wi-Fi to AWS IoT platform

Sensors

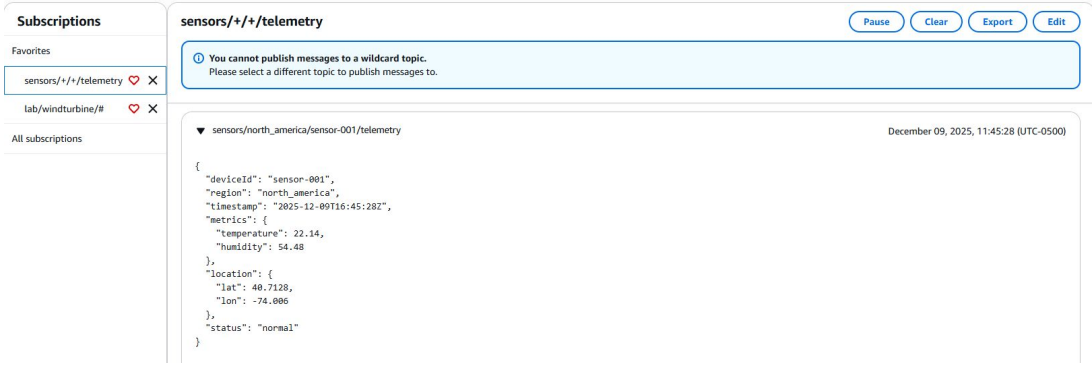
- SW420 Vibration Sensor
  - Based on threshold, detects High/Low values
  - Ability to adjust sensitivity



VCC → 3.3V on ESP32
GND → GND
Digital output (D0) → GPIO pin

```
C:\Users\harin\OneDrive\Desktop\geo-sensor-demo>python sensor_client.py
Connecting to a2ny46zv4dxt3b-ats.iot.us-east-1.amazonaws.com as sensor-001...
Connected!
Publishing: {"deviceId": "sensor-001", "region": "north_america", "timestamp": "2025-12-02T18:38:49Z", "metrics": {"temp
erature": 20.02, "humidity": 50.65}, "location": {"lat": 40.7128, "lon": -74.006}, "status": "normal"}
```

AWS IoT



1) **Connecting to AWS IoT:**
   - Setting up microcontroller to receive sensor data & using MQTT to publish/subscribe to models to implement telemetry

2) **Data Routing & Storing:**
   - Creating IoT rules to push telemetry to subscribed MQTT topics for storing sensor data

3) **Backend Processing:**
   - Using Lambda functions (ex: getDevices) to retrieve data and upload to web interface.


```
{
  "deviceId": "sensor-001",
  "region": "north_america",
  "timestamp": "2025-12-09T16:40:58Z",
  "metrics": {
    "temperature": 21.21,
    "humidity": 43.21
  },
  "location": {
    "lat": 40.7128,
    "lon": -74.006
  },
  "status": "normal"
}
```

Explanation of Code

- IQR is a data analysis method that reads a set of data and identifies the middle 50% of data points
- After identifying this median range, it will also identify any values outside of it and flag these as outlying data
- This method of analysis and categorization of data points are the crucial step to predicting failures before they happen

```
# 3. DETECT OUTLIERS USING IQR METHOD
# ==========================================
def detect_outliers_iqr(data, column,
multiplier=1.5):
    """
    Detect outliers using Interquartile Range
(IQR) method
    multiplier: typically 1.5 for outliers, 3.0
for extreme outliers
    """
    Q1 = data[column].quantile(0.25)
    Q3 = data[column].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - multiplier * IQR
    upper_bound = Q3 + multiplier * IQR

    outliers = (data[column] < lower_bound) |
(data[column] > upper_bound)

    return outliers, lower_bound, upper_bound

# Create a dataframe to store outlier flags
outlier_flags = pd.DataFrame(index=df.index)

# Dictionary to store outlier statistics
outlier_stats = {}

print("="*60)
print("OUTLIER DETECTION RESULTS")
print("="*60)

for col in sensor_columns:
    # Skip if column has too many missing values
```

Using Code for Failure Detection

- Developed code to read sensor data from a csv file, and using IQR read outlying data values
- After identifying outlying data, categorize affected sensors and components and suggest action items

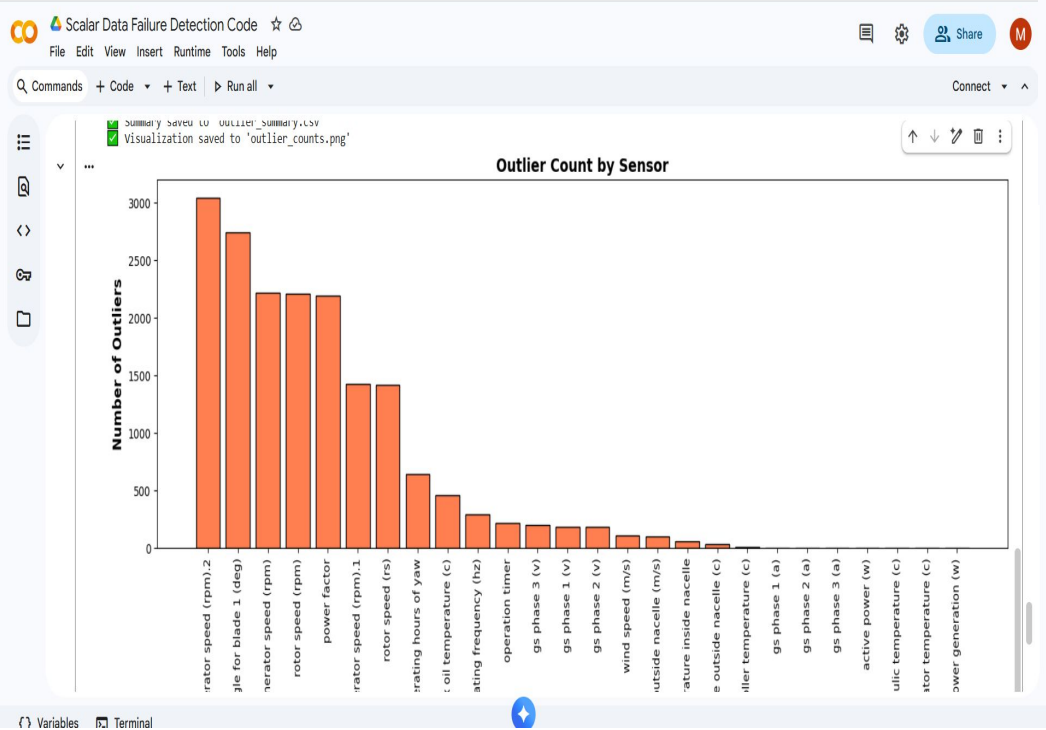
Scalar Data Failure Detection Code

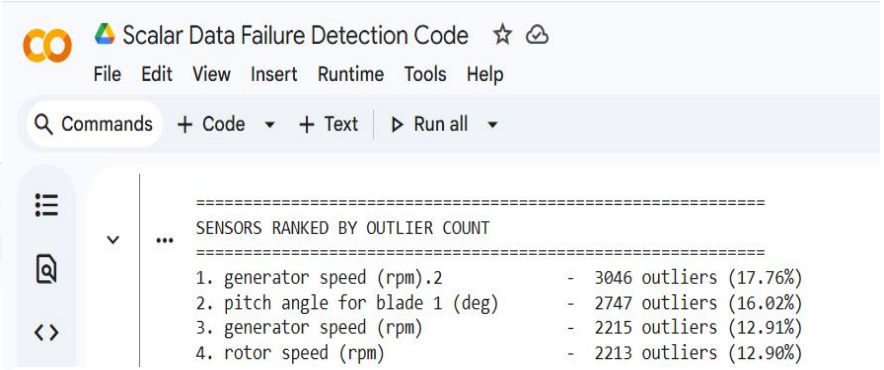
File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
============================================================
RECOMMENDED ACTION ITEMS
============================================================

Found 19 sensors with outliers. Recommended actions:


============================================================
🔧 GENERATOR SPEED (RPM).2
   Outliers: 3046 (17.76%)

   Action Items:
   1. 🚨 URGENT: Schedule immediate inspection
   2. Inspect mechanical connections and couplings
   3. Check for vibration or imbalance issues
   4. Verify sensor calibration and mounting

============================================================
🔧 PITCH ANGLE FOR BLADE 1 (DEG)
   Outliers: 2747 (16.02%)

   Action Items:
   1. 🚨 URGENT: Schedule immediate inspection
   2. Inspect pitch control system and actuators
   3. Check hydraulic pressure and fluid levels
   4. Verify pitch angle sensor calibration
   5. Examine blade bearings and pitch motors

============================================================
```

Outputs:





- These are important outputs when trying to determine affected components:
- **Generator Speed** - bearings, couplings, power inverter
- **Pitch Angle of Rotor Blade**- motors/actuators, hydraulics, sensors
- **Rotor Speed**- hub, blades, shafts, brake system

Extending Failure Detection Beyond IQR

- IQR code looks at which sensor appear abnormal
- Equation based code looks to see if physics between sensors is breaking
- Uses 6 turbine relationships (gearbox, pitch, generator, aero…)
- Produces labeled events: Healthy → Prefailure → Failure

```
============================================================
SENSORS RANKED BY OUTLIER COUNT
============================================================
1. generator speed (rpm).2              -  3046 outliers (17.76%)
2. pitch angle for blade 1 (deg)        -  2747 outliers (16.02%)
3. generator speed (rpm)                -  2215 outliers (12.91%)
4. rotor speed (rpm)                    -  2213 outliers (12.90%)
5. power factor                         -  2193 outliers (12.79%)
6. generator speed (rpm).1              -  1429 outliers ( 8.33%)
7. rotor speed (rs)                     -  1420 outliers ( 8.28%)
8. total operating hours of yaw         -   641 outliers ( 3.74%)
9. gearbox oil temperature (c)          -   465 outliers ( 2.71%)
10. operating frequency (hz)            -   297 outliers ( 1.73%)
11. operation timer                     -   217 outliers ( 1.27%)
12. gs phase 3 (v)                      -   205 outliers ( 1.20%)
13. gs phase 1 (v)                      -   188 outliers ( 1.10%)
14. gs phase 2 (v)                      -   185 outliers ( 1.08%)
15. wind speed (m/s)                    -   113 outliers ( 0.66%)
16. wind speed outside nacelle (m/s)    -   108 outliers ( 0.63%)
17. temperature inside nacelle          -    64 outliers ( 0.37%)
18. temperature outside nacelle (c)     -    37 outliers ( 0.22%)
19. controller temperature (c)          -    12 outliers ( 0.07%)
20. gs phase 1 (a)                      -     0 outliers ( 0.00%)
21. gs phase 2 (a)                      -     0 outliers ( 0.00%)
22. gs phase 3 (a)                      -     0 outliers ( 0.00%)
23. active power (w)                    -     0 outliers ( 0.00%)
24. hydraulic temperature (c)           -     0 outliers ( 0.00%)
25. generator stator temperature (c)    -     0 outliers ( 0.00%)
26. active power generation (w)         -     0 outliers ( 0.00%)
```

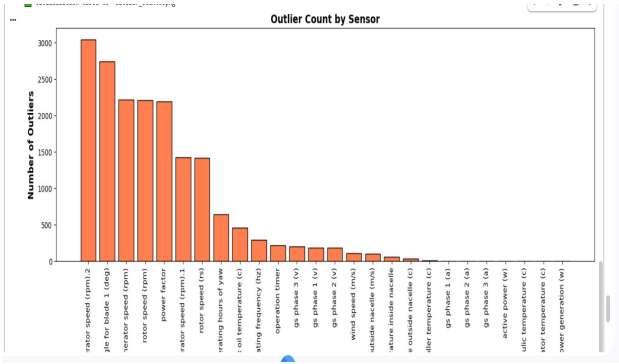
```
Equation applied: gear_oil_vs_rotor
Equation applied: stator_temp_vs_power
Equation applied: hydraulic_temp_vs_rotor
Equation applied: power_vs_wind_external
Equation applied: pitch1_vs_rotor
Equation applied: hydraulic_rotor_alt
```

How Equations Detect Failure

- Learn or use slope/intercept
- Track residual drift over time
- MAD gives stable “normal” baseline
- Thresholds → failure / prefailure

```
y_pred = slope * x + intercept
r = (y - y_pred).abs()

mad = np.median(r)
if mad == 0 or np.isnan(mad):
    std = r.std()
    if std == 0 or np.isnan(std):
        # no variation at all -> no events, zero score
        idx = df.index
        return (
            pd.Series(False, index=idx),
            pd.Series(False, index=idx),
            pd.Series(0.0, index=idx),
        )
    mad = std

roll_dev = r.rolling(ROLL_WINDOW, min_periods=1).mean()
score = roll_dev / (mad + 1e-9)  # normalized deviation
```

Integrating Both Approaches for Final System




```
Equation: Gear oil temp vs rotor speed (gear_oil_vs_rotor)
  Primary subsystem : Gearbox
  What it detects   : Overheating, lubrication failure
  Failures flagged  : 2406 rows (14.03% of dataset)
  Prefailures flags : 3196 rows (18.63% of dataset)

Equation: Power vs stator temperature (stator_temp_vs_power)
  Primary subsystem : Generator
  What it detects   : Cooling failure, winding heating
  Failures flagged  : 297 rows (1.73% of dataset)
  Prefailures flags : 1168 rows (6.81% of dataset)

Equation: Hydraulic temp vs rotor speed (hydraulic_temp_vs_rotor)
  Primary subsystem : Pitch Actuator
  What it detects   : Pitch system inefficiency, overheating
  Failures flagged  : 1435 rows (8.37% of dataset)
  Prefailures flags : 1522 rows (8.87% of dataset)
```

- IQR module detects raw sensor outliers
- Equation module detects behavior anomalies
- Together, they create:
  - Raw anomaly detection
  - Physics based failure detection
  - Probabilistic health score
- Final SCADA model will merge both → more accurate & earlier detection

How each of our contributions tie together

- **Quantitative Analysis (John):** Build mathematical relationships that explain failure behaviors.
- **Qualitative Analysis (Maritza):** Identify real-world failure themes from maintenance logs and OEM patterns.
- **FMEA (Katarina):** Integrate quantitative and qualitative results to rank turbine components by risk.
- **Sensors & IoT (Harinee):** Provide real-time turbine data through condition monitoring and AWS connectivity.
- **Data Processing (Sean & Michael):** Clean, classify, and detect abnormal patterns in the incoming data.
- **Prediction Modeling (Everyone):** Combine all insights to create a unified predictive-maintenance model linking failures, parts, maintenance, and troubleshooting.

Our next steps for this project include using this prediction model to focus on specific wind turbines. The RPN and risk levels for these turbines will be slightly different based on where they are located or if they are oriented differently. Also, to derive failure information for a wind turbine, you need to acquire its specific data over a period of time, through sensors, to provide a baseline data set.

Thank you
Any Questions?